- REPORT-

Student name: Nguyễn Minh Hoàng

Student ID: 26560020.

1. Experiment 2:

Listing 3

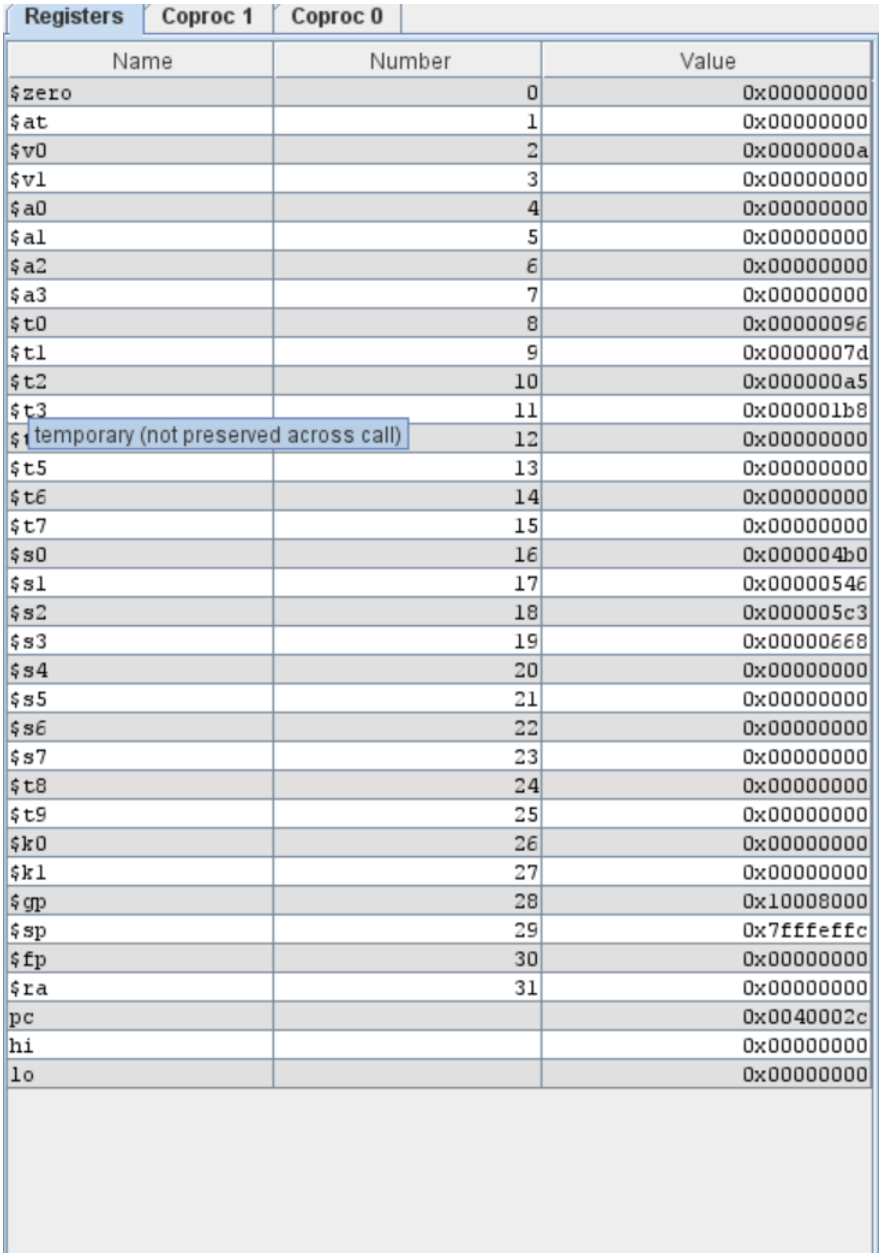
```
# Listing 3 - energy used by a household (fictitious readings,
kWh)
# Variable   Register
# reading0   $s0   start of month 1
# reading1   $s1   start of month 2
# reading2   $s2   start of month 3
# reading3   $s3   end of month 3
# usage1     $t0   energy used in month 1
# usage2     $t1   energy used in month 2
# usage3     $t2   energy used in month 3
# total      $t3   usage1 + usage2 + usage3
.text
.globl main
main:
    addi $s0, $zero, 1200
    addi $s1, $zero, 1350
    addi $s2, $zero, 1475
    addi $s3, $zero, 1640

    # >>> usage1 = reading1 - reading0
    # >>> usage2 = reading2 - reading1
    # >>> usage3 = reading3 - reading2
# >>> total = usage1 + usage2 + usage3   (two add instructions)
    sub  $t0, $s1, $s0
    sub  $t1, $s2, $s1
    sub  $t2, $s3, $s2
    add  $t3, $t0, $t1
    add  $t3, $t3, $t2

    li   $v0, 10
    Syscall
```

Check your result

With these readings, after the program has run, `$t0`, `$t1`, `$t2` and `$t3` must contain 150, 125, 165 and 440.



Name	Number	Value
$zero	0	0x00000000
$at	1	0x00000000
$v0	2	0x0000000a
$v1	3	0x00000000
$a0	4	0x00000000
$a1	5	0x00000000
$a2	6	0x00000000
$a3	7	0x00000000
$t0	8	0x00000096
$t1	9	0x0000007d
$t2	10	0x000000a5
$t3	11	0x000001b8
$t4	12	0x00000000
$t5	13	0x00000000
$t6	14	0x00000000
$t7	15	0x00000000
$s0	16	0x000004b0
$s1	17	0x00000546
$s2	18	0x000005c3
$s3	19	0x00000668
$s4	20	0x00000000
$s5	21	0x00000000
$s6	22	0x00000000
$s7	23	0x00000000
$t8	24	0x00000000
$t9	25	0x00000000
$k0	26	0x00000000
$k1	27	0x00000000
$gp	28	0x10008000
$sp	29	0x7fffeffc
$fp	30	0x00000000
$ra	31	0x00000000
pc		0x0040002c
hi		0x00000000
lo		0x00000000

Question 2.3 (Analyze) Calculate `reading3 - reading0` by hand and compare it with `total`. Why must the two numbers be equal? Explain in terms of what the meter readings represent.

Reading 3 - reading 0 = 1640 - 1200, which equals total. Therefore the energy used from the start of month 1 to the ends of month 3 are equals the sum of energy used during months 1,2 and 3.

Statement	Destination	Your MIPS instructions	PREDICT (decimal)	OBSERVE (decimal)
`x1 = y + z + w`	`$t0`	add $t0, $s0, $s1 add $t0, $t0, $s2	`16	16
`x2 = (y + z) - (w + 5)`	`$t1`	add $t5, $s0, $s1 addi $t1, $s2, 5 sub $t1, $t5, $t1	5	5
`x3 = y` (copy)	`$t2`	add $t2, $s0, $zero	8	8
`x4 = z - y`	`$t3`	Sub $t3, $s1, $s0	-3	-3

```
# Listing 4 - translating statements
# Register plan:
# y  = $s0 = 8
# z  = $s1 = 5
# w  = $s2 = 3
# x1 = $t0
# x2 = $t1
# x3 = $t2
# x4 = $t3
# $t5 is a temporary register

.text
.globl main
main:
    addi $s0, $zero, 8       # y = 8
    addi $s1, $zero, 5       # z = 5
    addi $s2, $zero, 3       # w = 3

    # >>> instructions for x1, x2, x3 and x4
        add  $t0, $s0, $s1  # x1 = y + z = 13
    add  $t0, $t0, $s2      # x1 = 13 + w = 16
```

```
    add  $t5, $s0, $s1     #temporary = y + z = 13
addi $t1, $s2, 5       # x2 = w + 5 = 8
    sub  $t1, $t5, $t1     # x2 = 13 - 8 =5

    add  $t2, $s0, $zero    # x3 = 8
    sub  $t3, $s1, $s0      # x4 = 5 - 8 = -3
    li   $v0, 10            # Exit program
    syscall
```

Question 2.4 (Analyze) The value of `x4` is negative. Show `$t3` in hexadecimal and in decimal. What do the leading hexadecimal digits of a negative number look like, and what does this suggest about how the processor stores negative numbers?

$t3 = 0xfffffffd in hexadecimal, which is -3 in decimal. The leading hexadecimal digits are F, suggesting that negative number stored using two's complement, with the most significant bit set to 1.